README

VisionEngine

Computer vision and LLM Vision for UI analysis and navigation graph building.

Overview

VisionEngine provides:

- **Analyzer interface** for screen analysis, element detection, text detection, and visual issue detection
- **NavigationGraph** for tracking app screen navigation with BFS pathfinding and DOT/JSON/Mermaid export
- **LLM Vision providers** for GPT-4o, Claude, Gemini, and Qwen-VL
- **OpenCV stubs** with build-tag gating for optional GoCV integration

Quick Start

```bash
# Run tests (no OpenCV required)
go test ./... -race -count=1

# Build
go build ./...

# With OpenCV support
go build -tags vision ./...
```

Packages

Package	Description
`pkg/analyzer`	Analyzer interface, types (UIElement, ScreenAnalysis, ScreenDiff, etc.)
`pkg/graph`	

Package	Description
	NavigationGraph with BFS pathfinding and DOT/JSON/Mermaid export
`pkg/llmvision`	VisionProvider interface with OpenAI, Anthropic, Gemini, Qwen adapters
`pkg/opencv`	OpenCV stub implementations (real impl behind `vision` build tag)
`pkg/config`	Configuration management via environment variables

NavigationGraph

The most important package -- imported by HelixQA for autonomous QA sessions.

```
g := graph.NewNavigationGraph()
g.AddScreen(analyzer.ScreenIdentity{ID: "home", Name: "Home"})
g.AddScreen(analyzer.ScreenIdentity{ID: "settings", Name:
        "Settings"})
g.AddTransition("home", "settings", analyzer.Action{Type:
        "click", Target: "gear"})
g.SetCurrent("home")

path, _ := g.PathTo("settings")
fmt.Println(graph.ExportMermaid(g))
```

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